PROJECT TITLE

APACE: Assistive Attendance Project Considering Efficiency

PRESENTORS:
ALVE, GABRIEL AARON L.
OREJOLA, CLAIRE L.

CS-2 PROJECT PRESENTATION

OBJECTIVES

Context:

The procedures for the current attendance system involve mainly manual work/processes. Though these processes of the attendance system make it functional, we noticed that:

Problems/Issues with current system/s:

- Manual attendance: Time-consuming for checking attendance. Slow, labor-intensive, prone to errors in records.
- QR-based Attendance: Can only handle one student per terminal, resulting in possible scalability issues for hundreds of students.

Obejctives:

- Allow teachers to record attendance sessions through exception-based inputs for student punctuality.
- Provide a transparent and assistive attendance tool that may help with the current limitations of traditional and semi-automated attendance systems.
- Allow students to view their attendance records for transparency of their recorded punctuality

PLANNED FEATURES

- Sign up, Login, Logout feature (Account Authentication)
- Role-based system attendance (Teachers, Students)
- Attendance features: Create Attendance, Attendance Dashboard.
- Data export of attendance records to CSV (Comma Separated Values)
- Pop-up messages for user actions in connection with event-driven GUI
- Attendance and Accessibility Settings

FEATURE IMPLEMENTATION

What? (The Feature)	Why? (Rationale)	How? (Implementation Steps)
Sign up, Login, Logout feature (Account Authentication)	To allow session handling of users while using the app, and additionally record/store the needed info for attendance marking. This additionally allows auditing/tracking of user activities in case of anonymous access to the app, and to allow data integrity inside the database through tracked info.	Sign up: Implement an account creation page that the user can use to make an account using their info, involving their name, username, account role type, grade level, and section (if applicable), and password. After validating data, store it in the appropriate database table. Login and Logout: Implement a login page asking the user to enter their account username and password. If the account username and password match (checked via a SQLite query for finding the account), store the authenticated name, surname, ID, and account role while logged in. Clear the session info upon logout.

FEATURE IMPLEMENTATION

What? (The Feature)	Why? (Rationale)	How? (Implementation Steps)
Role-based access control and attendance (Teachers, Students)	1. Prevent unauthorized access to certain app features. 2. To segregate the SQLite database schema for account types, which simplifies queries and reduces overhead effects for large numbers of users	In the SQLite database schema, add a separate table for teachers and students. Then, in the authentication logic service layer, after the signup data was validated, insert the data into its appropriate SQLite table depending on the account role. The account role can be chosen during sign-up Form. Upon login with the account, based on the account role, show only the appropriate features/content to avoid unauthorized feature access.

FEATURE IMPLEMENTATION

What? (The Feature)	Why? (Rationale)	How? (Implementation Steps)
Attendance features: Create Attendance, Attendance Dashboard.	Provides organized attendance management involving attendance sessions and records, specifically through the structured categories and the needed info of student attendance.	In the attendance dashboard, create a “Create Attendance” form that involves inputting the grade level and section, subject, date, and the start period relevant to the attendance. Additionally, in the attendance dashboard,

FEATURE IMPLEMENTATION

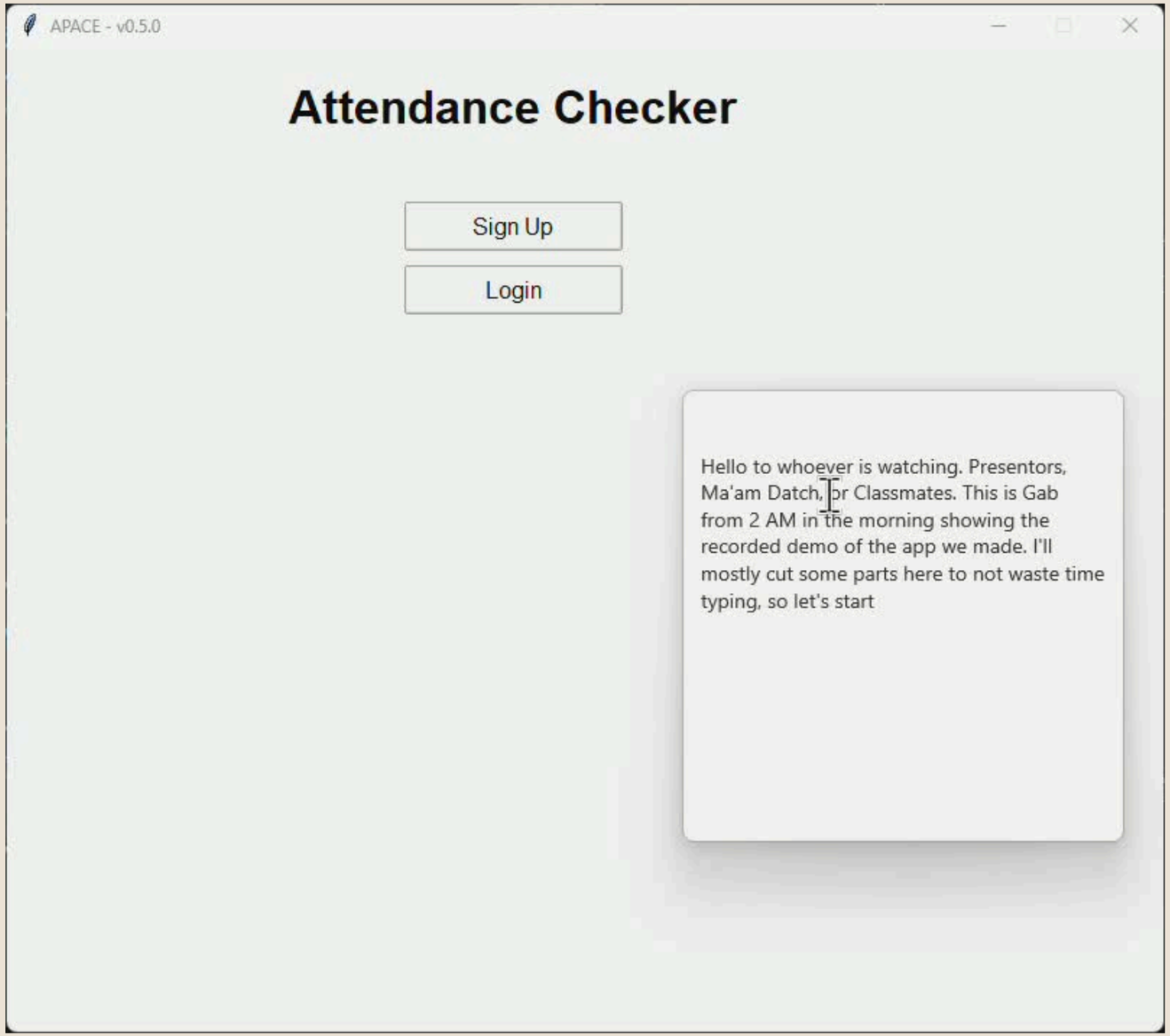
What? (The Feature)	Why? (Rationale)	How? (Implementation Steps)
Data export of attendance records to CSV (Comma Separated Values)	Allows data portability outside of the SQLite Database. It additionally allows compilation and ease of access, given that it is a CSV.	Use an SQLite query to fetch attendance sessions, attendance records, and relevant student info such that the attendance sessions. Then, use the pandas library to convert the query results into a CSV file.

FEATURE IMPLEMENTATION

What? (The Feature)	Why? (Rationale)	How? (Implementation Steps)
Pop-up messages for user actions in connection with event-driven GUI	Provide feedback and clarity on user actions, as well as inform the user of any errors in their inputs or actions and how to address them.	The UI Layer will read inputs and send them to the backend layer for input processing results. If the backend layer returns an error status of true, use a pop-up helper function and additionally use the returned error title and message. If there are no errors, give the user a pop-up of their successful user input (i.e, their input follows the happy path). Note: tkinter's messagebox was used for the popups. Moreover, the pop-up format mostly includes the type of error, what

FEATURE IMPLEMENTATION

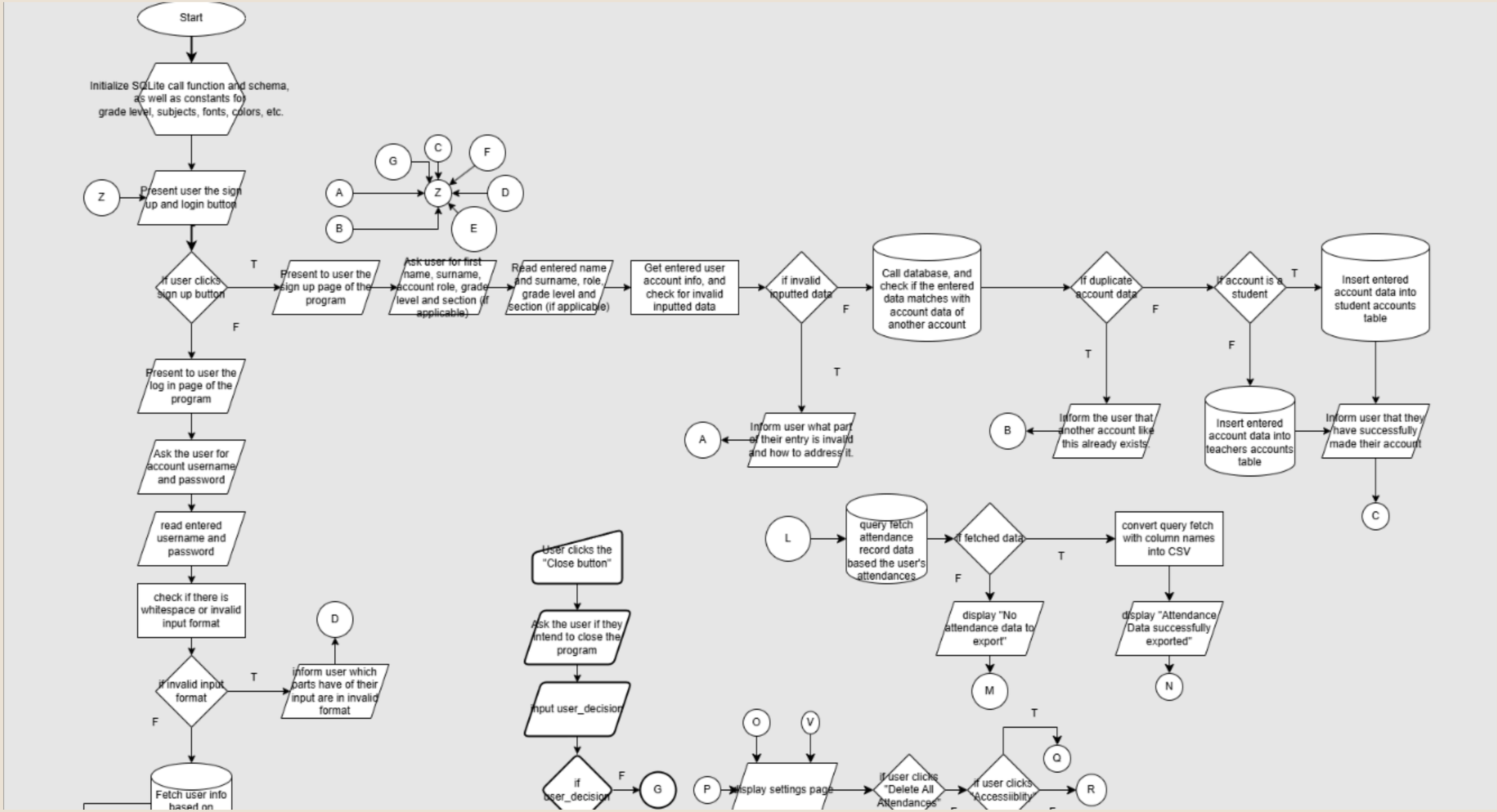
What? (The Feature)	Why? (Rationale)	How? (Implementation Steps)
System Settings (Attendance and Accessibility Settings)	Attendance Settings (Delete Attendance): To remove outdated/irrelevant attendance sessions and records Accessibility: To consider inclusion in UI design, and improve the usability of the app for those with disabilities	Delete All Attendances: Check first whether the user is a Teacher, as this is to prevent students from deleting attendance without notice. Then, ask the user to confirm their action. If they confirm their action, fetch the teacher's account ID, and delete all attendance sessions and records associated with the account ID. Accessibility: High Contrast Mode: Upon click, configure the app appearance such that the background does not blend in with the foreground/info needed by the user. Additionally, whenever the user's cursor hovers/selects an interactable widget, ensure that the focus color does not blend in with the info/widget.

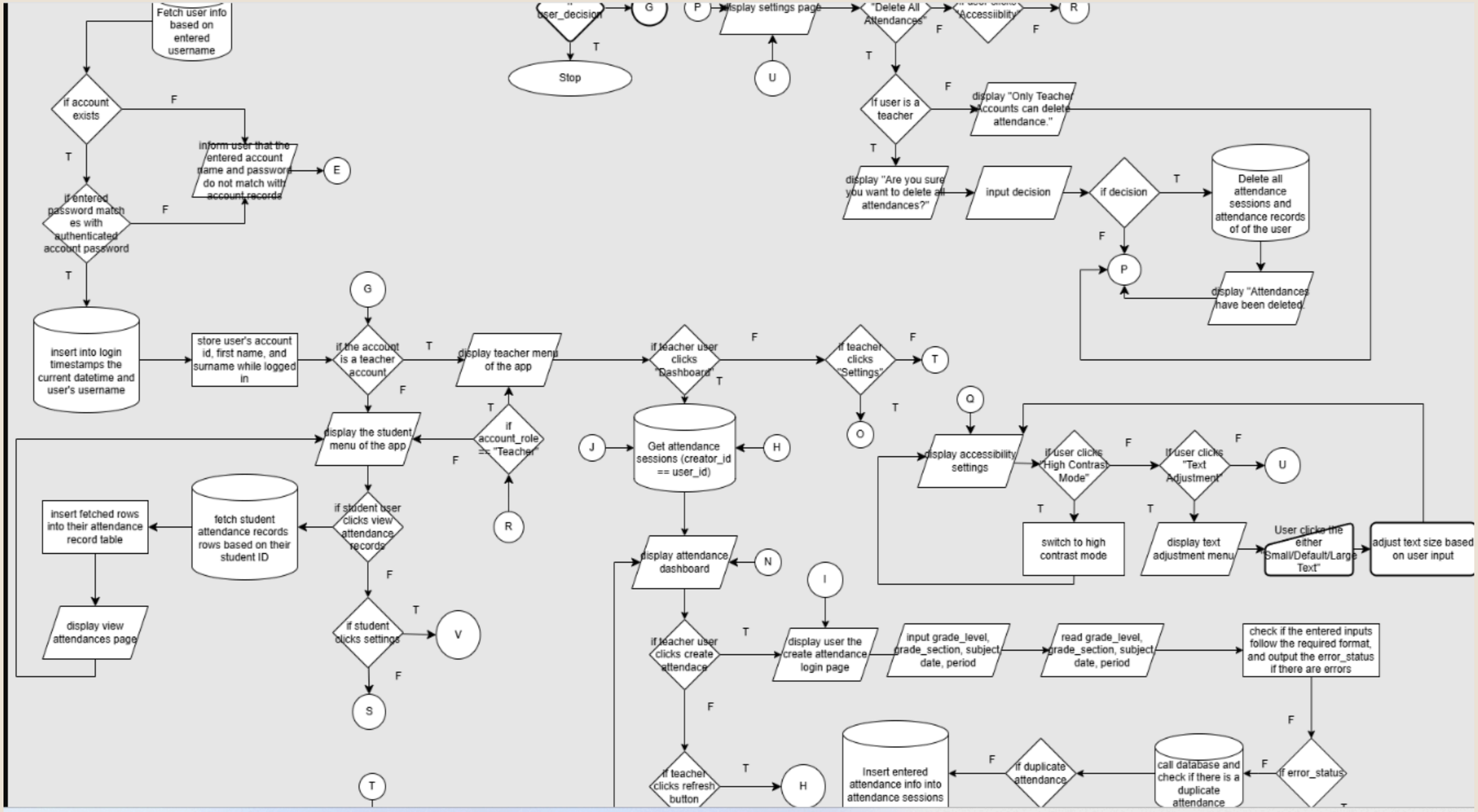


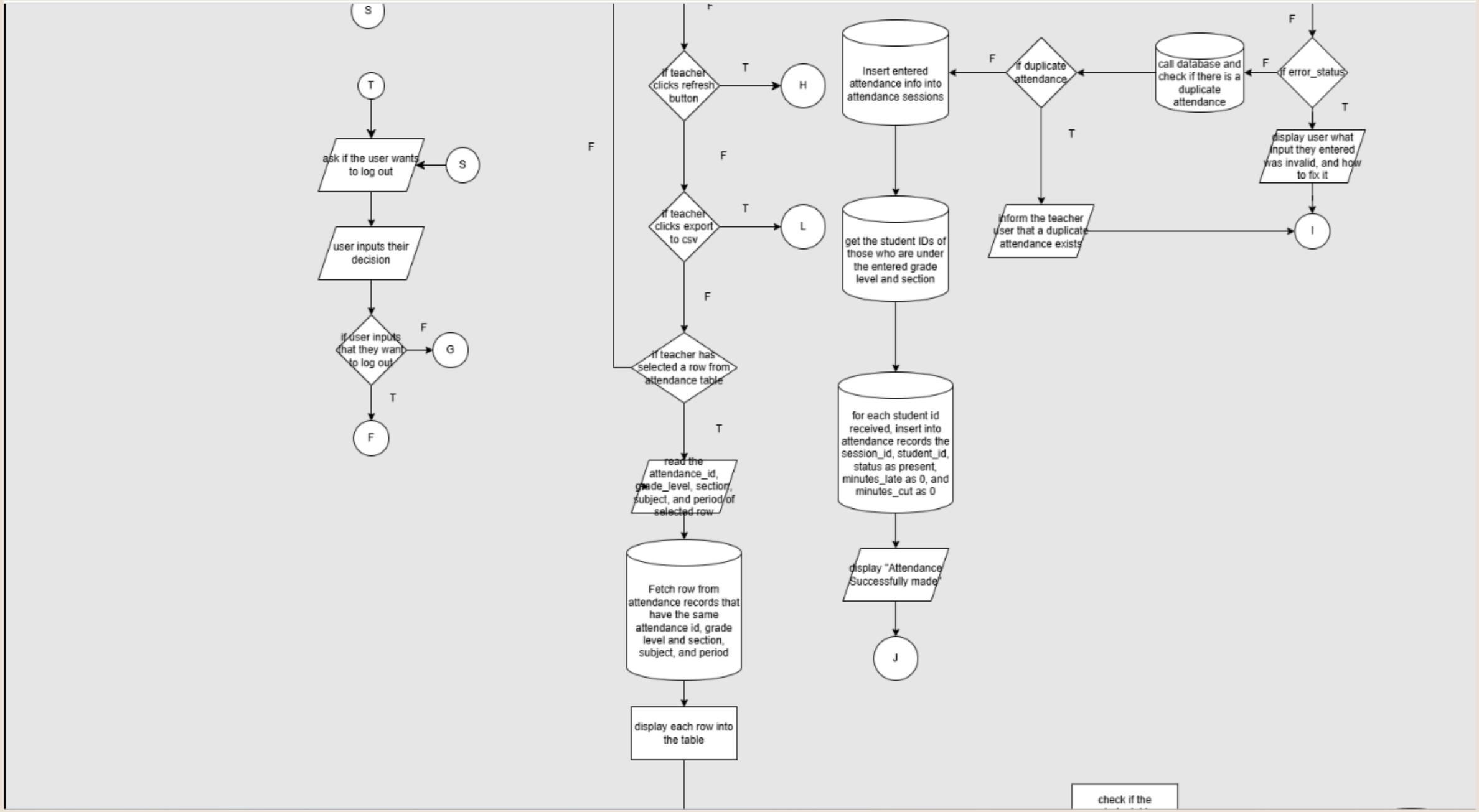
CS-2 PROJECT PRESENTATION

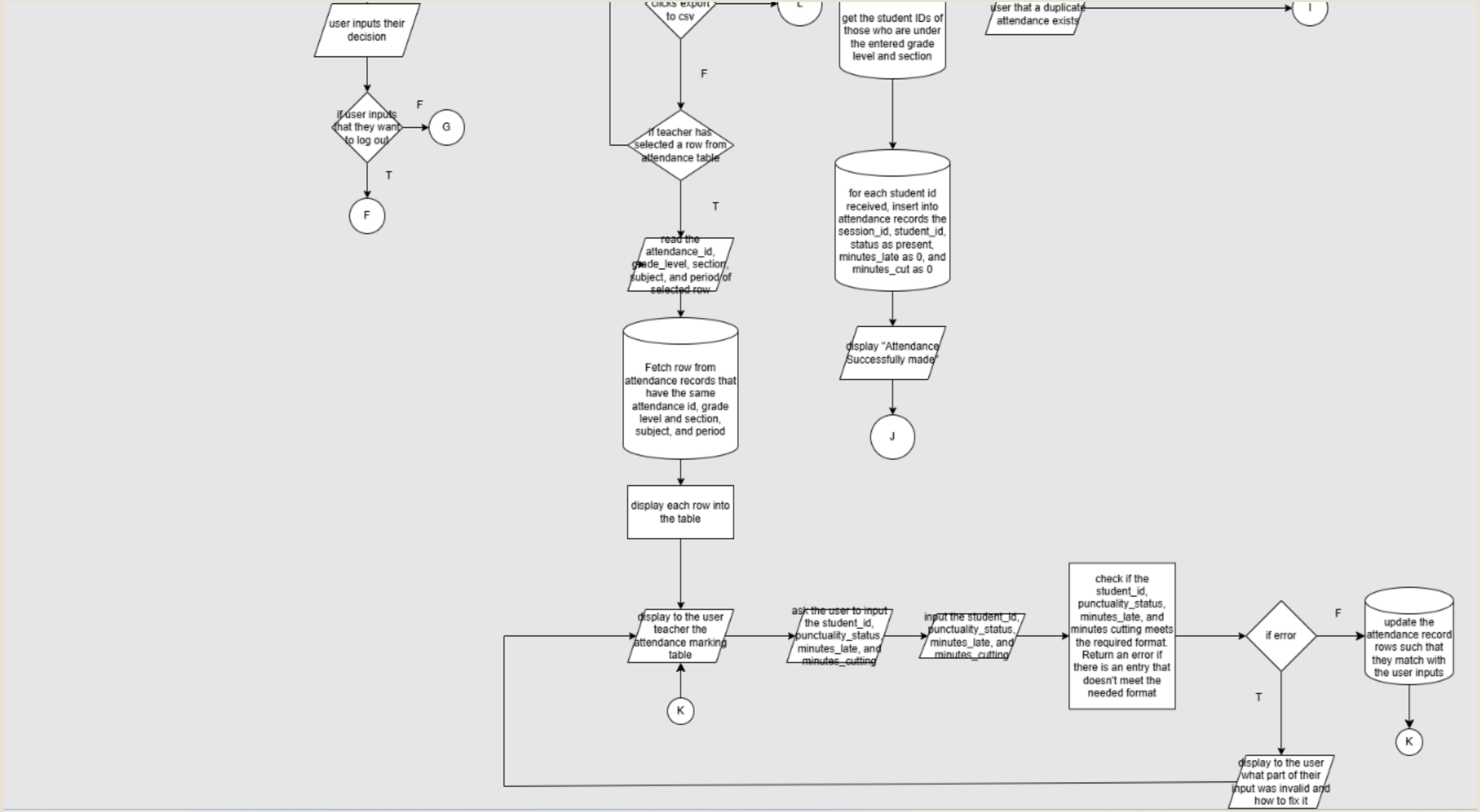
PROCESS FLOW

CS-2 PROJECT PRESENTATION


Start
Initialize SQLite call function and schema, as well as constants for grade level, subjects, fonts, colors, etc.
Z
Present user the sign up and login button
G
C
F
A
Z
D
B
E
If user clicks sign up button
T
F
Present to user the sign up page of the program
Ask user for first name, surname, account role, grade level and section (if applicable)
Read entered name and surname, role, grade level and section (if applicable)
Get entered user account info, and check for invalid inputted data
if invalid inputted data
F
T
Call database, and check if the entered data matches with account data of another account
If duplicate account data
F
T
If account is a student
T
F
Insert entered account data into student accounts table
Inform user that they have successfully made their account
C
Insert entered account data into teachers accounts table
Inform the user that another account like this already exists.
B
Inform user what part of their entry is invalid and how to address it.
A
Present to user the log in page of the program
Ask the user for account username and password
read entered username and password
check if there is whitespace or invalid input format
if invalid input format
T
F
inform user which parts have of their input are in invalid format
D
Fetch user info based on
User clicks the "Close button"
Ask the user if they intend to close the program
input user_decision
if user_decision
F
G
L
query fetch attendance record data based the user's attendances
if fetched data
T
F
convert query fetch with column names into CSV
display "No attendance data to export"
display "Attendance Data successfully exported"
M
N
O
V
P
display settings page
if user clicks "Delete All Attendances"
if user clicks "Accessiblity"
T
Q
R


Fetch user info based on entered username
if account exists
F
T
inform user that the entered account name and password do not match with account records
E
if entered password matches with authenticated account password
F
T
insert into login timestamps the current datetime and user's username
store user's account id, first name, and surname while logged in
G
if the account is a teacher account
T
F
display teacher menu of the app
if teacher user clicks "Dashboard"
F
T
if teacher clicks "Settings"
F
T
O
display the student menu of the app
T
if account_role == "Teacher"
F
R
J
Get attendance sessions (creator_id == user_id)
H
if student user clicks view attendance records
fetch student attendance records rows based on their student ID
insert fetched rows into their attendance record table
display view attendances page
F
if student clicks settings
T
V
F
S
display attendance dashboard
N
if teacher user clicks create attendace
T
F
display user the create attendance login page
input grade_level, grade_section, subject, date, period
read grade_level, grade_section, subject, date, period
check if the entered inputs follow the required format, and output the error_status if there are errors
if error_status
F
call database and check if there is a duplicate attendance
if duplicate attendance
F
Insert entered attendance info into attendance sessions
if teacher clicks refresh button
T
H
T
Stop
U
P
display settings page
"Delete All Attendances"
F
T
if user is a teacher
F
display "Only Teacher Accounts can delete attendance."
T
display "Are you sure you want to delete all attendances?"
input decision
if decision
T
F
Delete all attendance sessions and attendance records of of the user
P
display "Attendances have been deleted"
Q
display accessibility settings
if user clicks "High Contrast Mode"
F
T
switch to high contrast mode
if user clicks "Text Adjustment"
F
T
U
display text adjustment menu
User clicks the either "Small/Default/Large Text"
adjust text size based on user input
I
display user the create attendance login page


S
T
ask if the user wants to log out
S
user inputs their decision
if user inputs that they want to log out
F
G
T
F
F
if teacher clicks refresh button
T
H
F
if teacher clicks export to csv
T
L
F
if teacher has selected a row from attendance table
T
read the attendance_id, grade_level, section, subject, and period of selected row
Fetch row from attendance records that have the same attendance id, grade level and section, subject, and period
display each row into the table
Insert entered attendance info into attendance sessions
get the student IDs of those who are under the entered grade level and section
for each student id received, insert into attendance records the session_id, student_id, status as present, minutes_late as 0, and minutes_cut as 0
display "Attendance Successfully made"
J
F
if duplicate attendance
T
inform the teacher user that a duplicate attendance exists
I
call database and check if there is a duplicate attendance
F
if error_status
F
T
display user what input they entered was invalid, and how to fix it
check if the


user inputs their decision
if user inputs that they want to log out
F
G
T
F
clicks export to csv
L
F
if teacher has selected a row from attendance table
T
read the attendance_id, grade_level, section, subject, and period of selected row
Fetch row from attendance records that have the same attendance id, grade level and section, subject, and period
display each row into the table
display to the user teacher the attendance marking table
K
ask the user to input the student_id, punctuality_status, minutes_late, and minutes_cutting
input the student_id, punctuality_status, minutes_late, and minutes_cutting
check if the student_id, punctuality_status, minutes_late, and minutes cutting meets the required format. Return an error if there is an entry that doesn't meet the needed format
if error
F
update the attendance record rows such that they match with the user inputs
K
T
display to the user what part of their input was invalid and how to fix it
get the student IDs of those who are under the entered grade level and section
for each student id received, insert into attendance records the session_id, student_id, status as present, minutes_late as 0, and minutes_cut as 0
display "Attendance Successfully made"
J
user that a duplicate attendance exists
I

CODING INSIGHTS AND LESSONS LEARNED

- Lesson: Not all edge cases/bugs of a code program can be solved. Rather, some are more prioritized than others in terms of effects, scope, resources to address them, and more.
- There is no single best answer for any design choices of a coding project (some are more advantageous than others)
- When making features, it’s often helpful to first look at the scope, inputs, and outputs.

END OF PRESENTATION

APACE: Assistive Project Attendance Considering Efficiency

THANK YOU!

PRESENTORS:
ALVE, GABRIEL AARON L.
OREJOLA, CLAIRE L.

CS-2 PROJECT PRESENTATION